

# Quantum Computing — A Step-by-Step Tutorial

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# 1 Introduction: scope, conventions, and roadmap

Quantum computing is written in the language of *finite-dimensional linear algebra over the complex numbers*. This track builds exactly that language — the objects, operations, and theorems the postulates of quantum computing rely on — *before* we attach any physical interpretation. We stop at the bridge where “unit vector”, “unitary matrix”, and “projector” acquire the names “state”, “gate”, and “measurement”.

## 1.1 Scope and assumptions

- **Field.** Everything lives over the complex numbers  $\mathbb{C}$ . Real spaces appear only as a stepping stone.
- **Dimension.** *Finite-dimensional only:* every space is  $\mathbb{C}^n$  for some finite  $n \in \mathbb{N}$ . We never need infinite-dimensional Hilbert spaces, continuous spectra, or measure theory.
- **End goal.** A fluent command of:  $\mathbb{C}^n$ , inner products, Dirac notation, linear operators, the adjoint, Hermitian and unitary operators, the tensor product, and the spectral theorem. These are *precisely* the ingredients the postulates use — no more.
- **Rigor.** Every derivation and proof is shown step by step, at the level of finite-dimensional linear algebra (concrete matrices), not abstract operator theory.
- **Prerequisites.** Comfort with real vectors and matrices, and with basic complex arithmetic. Anything beyond that we re-derive as needed.

## 1.2 Notation and conventions

These conventions are fixed for the whole track; the preamble defines the macros so the notation is identical in every later section.

| Convention        | Symbol                            | Meaning (type / range)                                                   |
|-------------------|-----------------------------------|--------------------------------------------------------------------------|
| Imaginary unit    | $i$                               | $i^2 = -1$ . We write $i$ , never $j$ .                                  |
| Scalars           | $\alpha, \beta, z \in \mathbb{C}$ | complex numbers.                                                         |
| Complex conjugate | $\bar{z}$                         | for $z = a + ib$ with $a, b \in \mathbb{R}$ , $\bar{z} = a - ib$ .       |
| Vectors           | $v \in \mathbb{C}^n$              | a <i>column</i> vector, shape $n \times 1$ .                             |
| Index base        | $i \in \{0, \dots, n-1\}$         | <b>0-based</b> , to match qubit labels $ 0\rangle, \dots,  n-1\rangle$ . |
| Adjoint           | $A^\dagger$                       | conjugate transpose of $A$ (“ $A$ -dagger”).                             |
| Identity          | $\mathbb{1}$                      | the identity operator / matrix.                                          |

Two notation clashes, flagged now.

- **Conjugate.** The complex conjugate is written  $\bar{z}$  throughout. The physics literature often writes  $z^*$  for the same thing; treat  $\bar{z}$  and  $z^*$  as synonyms.
- **Indexing.** Pure-math linear algebra usually indexes from 1. We index from 0 so that basis vectors line up with computational-basis labels  $|0\rangle, |1\rangle, \dots$  later. Whenever an index appears, it runs over  $\{0, \dots, n-1\}$  unless stated otherwise.

### 1.3 Roadmap

Figure 1 is the whole plan on one page. Each box is a future step (one section, one file); an arrow  $A \rightarrow B$  means “ $A$  is used to define  $B$ ”. The chain is almost linear, which is why a strict step-by-step order works: each object is built only from objects already defined above it.

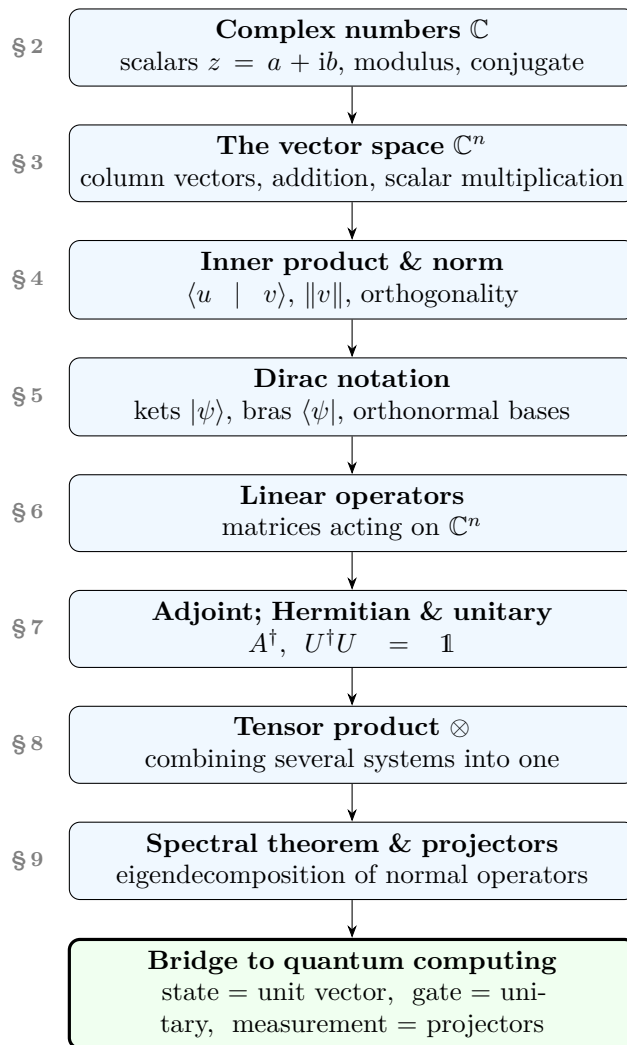


Figure 1: Roadmap for the math-first track. Every box is a forthcoming step; an arrow  $A \rightarrow B$  means “ $A$  is used to define  $B$ ”. We deliberately stop at the green bridge: once these objects exist, the postulates of quantum computing are one short step away.

The next step (§2) starts at the top of the chain: the complex numbers  $\mathbb{C}$  themselves — the scalars everything else is built from.

## 2 Complex numbers $\mathbb{C}$

The complex numbers are the scalars of quantum computing: every amplitude is a complex number, and the whole edifice of  $\mathbb{C}^n$ , inner products, and unitaries is built over  $\mathbb{C}$ . This step assembles the complex arithmetic we will lean on — in full, since the polar form and roots of unity reappear constantly later.

## 2.1 The field $\mathbb{C}$

**Definition 2.1** (Complex numbers).  $\mathbb{C} = \{a + ib : a, b \in \mathbb{R}\}$ , where  $i$  is a formal symbol with the single defining rule  $i^2 = -1$ . For  $z = a + ib$  we call  $a = \operatorname{Re} z \in \mathbb{R}$  the *real part* and  $b = \operatorname{Im} z \in \mathbb{R}$  the *imaginary part*.

**Flag.**  $\operatorname{Im} z$  is the *real number*  $b$ , not  $ib$ . So  $\operatorname{Im}(3 + 4i) = 4$ , a real number.

Addition is componentwise; multiplication is forced by distributivity together with  $i^2 = -1$ :

$$(a + ib) + (c + id) = (a + c) + i(b + d), \quad (1)$$

$$\begin{aligned} (a + ib)(c + id) &= ac + iad + ibc + i^2 bd && \text{(distribute)} \\ &= (ac - bd) + i(ad + bc). && \text{(use } i^2 = -1) \end{aligned} \quad (2)$$

With these operations  $\mathbb{C}$  is a *field*: addition and multiplication are commutative and associative, distribute, have identities 0 and 1, and every  $z \neq 0$  has a multiplicative inverse (constructed in section 2.4).

## 2.2 The complex plane

Identifying  $z = a + ib$  with the point  $(a, b) \in \mathbb{R}^2$  gives the geometric picture in fig. 2:  $\mathbb{C}$  is a plane, with the real numbers on the horizontal axis and the multiples of  $i$  on the vertical axis.

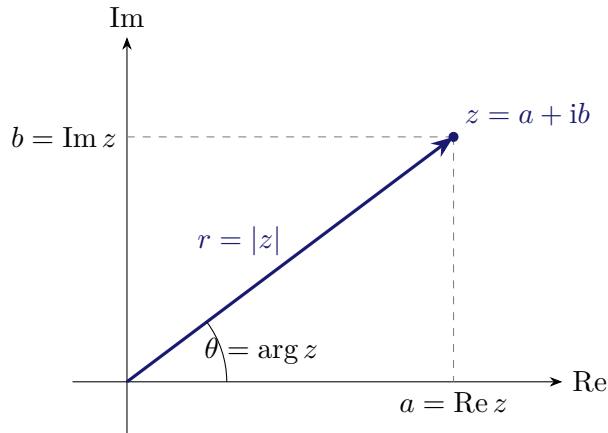


Figure 2: The complex plane. A number  $z = a + ib$  is the point  $(a, b)$ , equivalently the arrow from the origin of length  $r = |z| = \sqrt{a^2 + b^2}$  making angle  $\theta = \arg z$  with the positive real axis. Cartesian form  $(a, b)$  and polar form  $(r, \theta)$  are two coordinates for the same point.

## 2.3 Conjugate and modulus

**Definition 2.2** (Conjugate, modulus). The *complex conjugate* of  $z = a + ib$  is  $\bar{z} = a - ib$  (reflection across the real axis). The *modulus* (absolute value) is  $|z| = \sqrt{a^2 + b^2} \in \mathbb{R}_{\geq 0}$ , the length of the arrow in fig. 2.

The single most useful identity links the two:

$$z \bar{z} = (a + ib)(a - ib) = a^2 - (ib)^2 = a^2 + b^2 = |z|^2. \quad (3)$$

So  $|z| = \sqrt{z \bar{z}}$ , and  $z \bar{z}$  is always a nonnegative *real* number. The core algebraic properties:

| Property             | Statement                                                                                             | Why                            |
|----------------------|-------------------------------------------------------------------------------------------------------|--------------------------------|
| conjugate of sum     | $\overline{z + w} = \overline{z} + \overline{w}$                                                      | flip sign of each Im part      |
| conjugate of product | $\overline{zw} = \overline{z} \overline{w}$                                                           | expand both sides (below)      |
| involution           | $\overline{\overline{z}} = z$                                                                         | flip the sign twice            |
| real/imag recovery   | $\operatorname{Re} z = \frac{z + \overline{z}}{2}, \operatorname{Im} z = \frac{z - \overline{z}}{2i}$ | add/subtract $z, \overline{z}$ |
| modulus multiplies   | $ zw  =  z   w $                                                                                      | proof below                    |
| triangle inequality  | $ z + w  \leq  z  +  w $                                                                              | lengths of a vector sum        |

*Proof that  $|zw| = |z||w|$ .* Work with squared moduli (both sides nonnegative, so squaring is reversible):

$$|zw|^2 = (zw) \overline{(zw)} \quad \text{by (3)} \quad (4)$$

$$= z w \overline{z} \overline{w} \quad \overline{zw} = \overline{z} \overline{w} \quad (5)$$

$$= (z\overline{z})(w\overline{w}) \quad \text{commute factors} \quad (6)$$

$$= |z|^2 |w|^2. \quad \text{by (3)} \quad (7)$$

Take nonnegative square roots:  $|zw| = |z||w|$ . □

## 2.4 Reciprocal and division

For  $z \neq 0$  we have  $|z|^2 > 0$ , so we may divide by it. From (3),  $z \cdot \frac{\overline{z}}{|z|^2} = \frac{z\overline{z}}{|z|^2} = 1$ , hence

$$z^{-1} = \frac{\overline{z}}{|z|^2}, \quad \text{and division is} \quad \frac{w}{z} = w z^{-1} = \frac{w \overline{z}}{|z|^2}. \quad (8)$$

The trick “multiply top and bottom by the conjugate of the denominator” is exactly this formula.

## 2.5 Polar form and Euler’s formula

Reading  $(r, \theta)$  off fig. 2 with  $r = |z|$  and  $\theta = \arg z$  (the angle, defined modulo  $2\pi$ ) gives  $a = r \cos \theta$ ,  $b = r \sin \theta$ , so

$$z = r(\cos \theta + i \sin \theta). \quad (9)$$

The bracket has a name. Expand  $e^{i\theta}$  as the exponential power series and split into even/odd powers of  $\theta$ , using the 4-cycle  $i^0 = 1$ ,  $i^1 = i$ ,  $i^2 = -1$ ,  $i^3 = -i$ :

$$e^{i\theta} = \sum_{k=0}^{\infty} \frac{(i\theta)^k}{k!} \quad \text{Maclaurin series of exp} \quad (10)$$

$$= \sum_{m=0}^{\infty} \frac{(i\theta)^{2m}}{(2m)!} + \sum_{m=0}^{\infty} \frac{(i\theta)^{2m+1}}{(2m+1)!} \quad \text{split even } (k=2m) / \text{ odd } (k=2m+1) \quad (11)$$

$$= \sum_{m=0}^{\infty} \frac{(-1)^m \theta^{2m}}{(2m)!} + i \sum_{m=0}^{\infty} \frac{(-1)^m \theta^{2m+1}}{(2m+1)!} \quad i^{2m} = (-1)^m, \quad i^{2m+1} = i(-1)^m \quad (12)$$

$$= \cos \theta + i \sin \theta. \quad \text{Maclaurin series of cos, sin} \quad (13)$$

(The rearrangement is legal because the exp series converges absolutely.) This is **Euler’s formula**, giving the compact polar form

$$\boxed{z = r e^{i\theta}}, \quad r = |z| \geq 0, \quad \theta = \arg z. \quad (14)$$

**Multiplication = scale and rotate.** In polar form, multiplication is transparent. Using the trig addition formulas,

$$\begin{aligned} (\cos \alpha + i \sin \alpha)(\cos \beta + i \sin \beta) &= (\cos \alpha \cos \beta - \sin \alpha \sin \beta) \\ &\quad + i(\sin \alpha \cos \beta + \cos \alpha \sin \beta) \end{aligned} \quad (15)$$

$$= \cos(\alpha + \beta) + i \sin(\alpha + \beta), \quad (16)$$

i.e.  $e^{i\alpha}e^{i\beta} = e^{i(\alpha+\beta)}$ . Therefore

$$(r_1 e^{i\theta_1})(r_2 e^{i\theta_2}) = r_1 r_2 e^{i(\theta_1+\theta_2)} : \quad (17)$$

*multiply the moduli, add the angles.* Multiplying by  $e^{i\theta}$  (modulus 1) is a pure rotation by  $\theta$  — the seed of every quantum gate that is a rotation.

## 2.6 Roots of unity

The equation  $z^n = 1$  has exactly  $n$  solutions, the  $n$ -th roots of unity

$$\omega_k = e^{2\pi i k/n}, \quad k \in \{0, \dots, n-1\}, \quad (18)$$

since  $\omega_k^n = e^{2\pi i k} = 1$ . They sit equally spaced on the unit circle (fig. 3). They are the backbone of the quantum Fourier transform, where these equal phase steps create the interference pattern that makes algorithms like Shor's work.

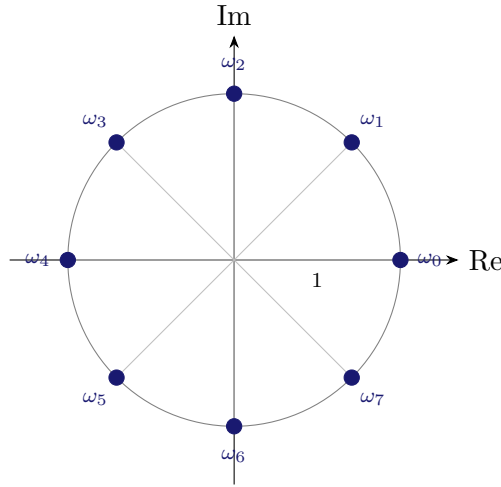


Figure 3: The  $n = 8$  roots of unity:  $\omega_k = e^{2\pi i k/8}$  for  $k = 0, \dots, 7$ , equally spaced on the unit circle at angle steps of  $2\pi/8 = 45^\circ$ , starting from  $\omega_0 = 1$ . Each satisfies  $\omega_k^8 = 1$ . These equally-spaced phases are the engine of the quantum Fourier transform (a later track).

## 2.7 Worked example

**Example 2.1.** Take  $z = 1 + i$  (so  $a = b = 1$ ). Compute its invariants, then double-check  $z^2$  two ways.

$$\operatorname{Re} z = 1, \quad \operatorname{Im} z = 1, \quad \bar{z} = 1 - i. \quad (19)$$

$$|z| = \sqrt{1^2 + 1^2} = \sqrt{2}. \quad (20)$$

$$\arg z = \arctan \frac{1}{1} = \frac{\pi}{4} \quad (\text{first quadrant}), \quad \text{so} \quad z = \sqrt{2} e^{i\pi/4}. \quad (21)$$

$$z^{-1} = \frac{\bar{z}}{|z|^2} = \frac{1-i}{2} \quad \left[ \text{check: } z z^{-1} = \frac{(1+i)(1-i)}{2} = \frac{1+1}{2} = 1. \checkmark \right] \quad (22)$$

Now  $z^2$ , computed both ways as a cross-check:

$$\text{rectangular: } z^2 = (1 + i)^2 = 1 + 2i + i^2 = 2i, \quad (23)$$

$$\text{polar: } z^2 = (\sqrt{2} e^{i\pi/4})^2 = 2 e^{i\pi/2} = 2(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2}) = 2i. \quad \checkmark \quad (24)$$

Both routes agree: squaring scaled the modulus  $\sqrt{2} \mapsto 2$  and doubled the angle  $\frac{\pi}{4} \mapsto \frac{\pi}{2}$ , exactly as “scale and rotate” predicts.

## 2.8 Why this matters for quantum computing

**Remark 2.1.** Three forward-pointers, so the motivation is on record:

- **Amplitudes are complex.** A qubit state carries complex amplitudes; adding them produces *interference*, the resource quantum algorithms exploit.
- **Phase is physical.** A factor  $e^{i\theta}$  (modulus 1) is a *phase*. *Relative* phases between amplitudes are observable; a single *global* phase  $e^{i\theta} |\psi\rangle$  is not — we will make this precise once states are defined.
- **Roots of unity drive the QFT.** The equally-spaced phases  $\omega_k$  are exactly the entries of the quantum Fourier transform matrix.

Next (§3): assemble these scalars into the vector space  $\mathbb{C}^n$  — the space a quantum state actually lives in.

## 3 The vector space $\mathbb{C}^n$

A single quantum state is a vector in  $\mathbb{C}^n$ . This step builds that space: its elements, its two operations, its standard basis, and the notions of independence, basis, and dimension — everything we need before adding an inner product (§4).

### 3.1 Definition

**Definition 3.1** (The space  $\mathbb{C}^n$ ). For  $n \in \mathbb{N}$ ,  $\mathbb{C}^n$  is the set of *column vectors* with  $n$  complex entries,

$$v = \begin{pmatrix} v_0 \\ v_1 \\ \vdots \\ v_{n-1} \end{pmatrix}, \quad v_i \in \mathbb{C}, \quad i \in \{0, \dots, n-1\}.$$

The entry  $v_i \in \mathbb{C}$  is the  $i$ -th *coordinate* of  $v$ . The shape of  $v$  is  $n \times 1$ .



**Flag (notation).**  $v_i$  (subscript) is a *scalar*, the  $i$ -th coordinate of the single vector  $v$ . Later we also index whole vectors, e.g.  $v^{(1)}, v^{(2)}$ ; we use superscripts in parentheses for that, to avoid clashing with coordinates.

### 3.2 The two operations

Addition and scalar multiplication are *componentwise*: for  $u, v \in \mathbb{C}^n$  and  $\alpha \in \mathbb{C}$ ,

$$(u + v)_i = u_i + v_i, \quad (\alpha v)_i = \alpha v_i, \quad i \in \{0, \dots, n-1\}. \quad (25)$$

With these,  $\mathbb{C}^n$  is a *complex vector space*: the eight axioms below all hold, each inherited entry-by-entry from the field  $\mathbb{C}$  of §2. Write  $\mathbf{0}$  for the zero vector (all entries 0) and  $-v$  for  $(-1)v$ .

| Axiom                       | Statement                                |
|-----------------------------|------------------------------------------|
| commutativity (+)           | $u + v = v + u$                          |
| associativity (+)           | $(u + v) + w = u + (v + w)$              |
| additive identity           | $v + \mathbf{0} = v$                     |
| additive inverse            | $v + (-v) = \mathbf{0}$                  |
| scalar identity             | $1 v = v$                                |
| scalar associativity        | $(\alpha\beta)v = \alpha(\beta v)$       |
| distributivity over vectors | $\alpha(u + v) = \alpha u + \alpha v$    |
| distributivity over scalars | $(\alpha + \beta)v = \alpha v + \beta v$ |

**Flag.  $\mathbf{0}$**  (bold, the zero *vector*) is not the scalar  $0 \in \mathbb{C}$ ; and the basis vector  $e_i$  below is unrelated to Euler's number  $e$  of §2.

### 3.3 Standard basis, span, independence, dimension

**Definition 3.2** (Standard basis). The *standard basis* (also called the *computational basis*) of  $\mathbb{C}^n$  is the set  $\{e_0, \dots, e_{n-1}\}$ . Here  $e_i$  has a 1 in coordinate  $i$  and a 0 in every other coordinate:

$$e_0 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad e_1 = \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \quad \dots, \quad e_{n-1} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}.$$

Every  $v \in \mathbb{C}^n$  decomposes along this basis, as in fig. 4:

$$v = \sum_{i=0}^{n-1} v_i e_i. \quad (26)$$

**Definition 3.3** (Span, linear independence, basis, dimension). A *linear combination* of  $v^{(1)}, \dots, v^{(k)}$  is any  $\sum_j c_j v^{(j)}$  with  $c_j \in \mathbb{C}$ . Their *span* is the set of all such combinations. The vectors are *linearly independent* if

$$\sum_j c_j v^{(j)} = \mathbf{0} \implies c_1 = \dots = c_k = 0 \quad (\text{the only way to reach } \mathbf{0} \text{ is all } c_j = 0).$$

A *basis* is a linearly independent spanning set; the *dimension* is the number of vectors in any basis.

$$\begin{array}{c}
\text{index } 0 \\
\text{index } 1 \\
\vdots \\
\text{index } n-1
\end{array}
\begin{pmatrix} v_0 \\ v_1 \\ \vdots \\ v_{n-1} \end{pmatrix}
\left. \vphantom{\begin{pmatrix} v_0 \\ v_1 \\ \vdots \\ v_{n-1} \end{pmatrix}} \right\} \begin{array}{l} n \text{ complex entries,} \\ \text{each } v_i \in \mathbb{C} \end{array}$$

$$v = \sum_{i=0}^{n-1} v_i e_i = v_0 e_0 + v_1 e_1 + \cdots + v_{n-1} e_{n-1}$$

Figure 4: A vector  $v \in \mathbb{C}^n$  is a column of  $n$  complex numbers, indexed  $0, \dots, n-1$ . Its entries  $v_i \in \mathbb{C}$  are its *coordinates* in the standard basis  $\{e_0, \dots, e_{n-1}\}$ , and  $v$  is the unique linear combination  $\sum_i v_i e_i$ .

**Theorem 3.1** (Coordinates are unique;  $\dim_{\mathbb{C}} \mathbb{C}^n = n$ ).  *$\{e_0, \dots, e_{n-1}\}$  is a basis of  $\mathbb{C}^n$ , and the coordinates in (26) are unique. Hence every basis of  $\mathbb{C}^n$  has exactly  $n$  elements, i.e.  $\dim_{\mathbb{C}} \mathbb{C}^n = n$ .*

*Proof. Spanning:* (26) writes an arbitrary  $v$  as a combination of the  $e_i$ . *Independence:* if  $\sum_i c_i e_i = \mathbf{0}$ , then reading off coordinate  $i$  of the left side gives  $c_i = 0$  for every  $i$  (the  $i$ -th coordinate of  $\sum_i c_i e_i$  is exactly  $c_i$ ). *Uniqueness of coordinates:* if  $\sum_i v_i e_i = \sum_i v'_i e_i$ , subtract to get  $\sum_i (v_i - v'_i) e_i = \mathbf{0}$ ; independence forces  $v_i = v'_i$  for all  $i$ . That  $\{e_i\}$  has  $n$  elements gives  $\dim_{\mathbb{C}} \mathbb{C}^n = n$ ; that *every* basis has the same size  $n$  is the standard exchange (Steinitz) theorem of finite-dimensional linear algebra, which we take as known.  $\square$

### 3.4 The case $\mathbb{C}^2$ : one qubit's worth of space

The space of a single qubit (before normalization) is  $\mathbb{C}^2$ , with standard basis

$$e_0 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad e_1 = \begin{pmatrix} 0 \\ 1 \end{pmatrix},$$

and a general vector

$$v = \alpha e_0 + \beta e_1 = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}, \quad \alpha, \beta \in \mathbb{C}. \quad (27)$$

In Dirac notation (§5) we will rename  $e_0, e_1$  to  $|0\rangle, |1\rangle$ , so this is the familiar  $\alpha|0\rangle + \beta|1\rangle$  — but that is just relabelling the basis vectors we already have.

**Remark 3.1** (Real vs. complex dimension — flag). As a *complex* vector space,  $\dim_{\mathbb{C}} \mathbb{C}^2 = 2$ . As a *real* vector space (treating each complex entry as a real pair  $a + ib$ ),  $\dim_{\mathbb{R}} \mathbb{C}^2 = 4$ . The qubit's continuous parameters come from this real count, but its linear algebra is governed by the complex count. We always mean  $\dim_{\mathbb{C}}$  unless we write  $\dim_{\mathbb{R}}$ .

**Remark 3.2** (Why dimension is the headline number).  $n$  qubits share a *single* state vector in  $\mathbb{C}^{2^n}$  (built with the tensor product, §8). The dimension is  $2^n$  — exponential in the number of qubits. That exponential state space is where quantum computing's potential advantage lives.

### 3.5 Worked example

**Example 3.1.** In  $\mathbb{C}^2$  take  $v = \begin{pmatrix} 1+i \\ 2 \end{pmatrix}$  and  $w = \begin{pmatrix} 0 \\ i \end{pmatrix}$ .

**Decomposition and operations.**

$$v = (1+i)e_0 + 2e_1, \quad (28)$$

$$2v = \begin{pmatrix} 2+2i \\ 4 \end{pmatrix}, \quad v+w = \begin{pmatrix} 1+i \\ 2+i \end{pmatrix}. \quad (29)$$

**Are  $v, w$  a basis of  $\mathbb{C}^2$ ?** They are two vectors in a 2-dimensional space, so it suffices to check independence. Place them as columns of a matrix and take the determinant:

$$\det \begin{pmatrix} 1+i & 0 \\ 2 & i \end{pmatrix} = (1+i)i - 0 \cdot 2 = i + i^2 = -1 + i \neq 0. \quad (30)$$

A nonzero determinant means the columns are linearly independent; two independent vectors in  $\mathbb{C}^2$  form a basis. ✓

Next (§4): equip  $\mathbb{C}^n$  with an *inner product*, which gives lengths, angles, and orthogonality — the structure that turns a bare vector space into the geometry quantum mechanics uses.

## 4 Inner product, norm, and orthogonality

A bare vector space has no notion of length or angle. The *inner product* adds exactly that geometry: it gives norms, orthogonality, and the projection formula that, in §5 and beyond, becomes “amplitude” and “measurement probability”. We work on  $\mathbb{C}^n$  throughout.

### 4.1 The standard (Hermitian) inner product

**Definition 4.1** (Inner product on  $\mathbb{C}^n$ ). For  $u, v \in \mathbb{C}^n$ ,

$$\langle u | v \rangle = \sum_{i=0}^{n-1} \overline{u_i} v_i \in \mathbb{C}. \quad (31)$$

The first vector is conjugated, the second is not.

**Convention clash — flagged and committed.** Two equally common conventions exist:

- **Physics** (ours): conjugate the *first* slot, so  $\langle u | v \rangle$  is *linear in the second* argument and *conjugate-linear in the first*. This matches Dirac notation  $\langle \psi | \phi \rangle$  in §5.
- **Many math texts**: conjugate the *second* slot instead.

These differ only by an overall conjugation, but they negate which slot is linear. **We fix the physics convention (31) for the entire track.**

## 4.2 The four defining properties

Each follows directly from (31) and the conjugate algebra of §2.

**Theorem 4.1** (Inner-product axioms). *For all  $u, v, w \in \mathbb{C}^n$  and  $\alpha, \beta \in \mathbb{C}$ :*

$$(i) \text{ conjugate symmetry: } \langle u | v \rangle = \overline{\langle v | u \rangle}, \quad (32)$$

$$(ii) \text{ linear in 2nd arg: } \langle u | \alpha v + \beta w \rangle = \alpha \langle u | v \rangle + \beta \langle u | w \rangle, \quad (33)$$

$$(iii) \text{ conj.-linear in 1st arg: } \langle \alpha u + \beta w | v \rangle = \overline{\alpha} \langle u | v \rangle + \overline{\beta} \langle w | v \rangle, \quad (34)$$

$$(iv) \text{ positive-definite: } \langle v | v \rangle \in \mathbb{R}_{\geq 0}, \quad \langle v | v \rangle = 0 \iff v = \mathbf{0}. \quad (35)$$

*Proof.* (i) Using  $\overline{\sum} = \sum$  - and  $\overline{v_i u_i} = v_i \overline{u_i}$ ,

$$\overline{\langle v | u \rangle} = \overline{\sum_i \overline{v_i} u_i} = \sum_i \overline{\overline{v_i} u_i} = \sum_i v_i \overline{u_i} = \sum_i \overline{u_i} v_i = \langle u | v \rangle.$$

(ii) Linearity of the sum:  $\langle u | \alpha v + \beta w \rangle = \sum_i \overline{u_i} (\alpha v_i + \beta w_i) = \alpha \sum_i \overline{u_i} v_i + \beta \sum_i \overline{u_i} w_i$ . (iii) Combine (i) and (ii):  $\langle \alpha u + \beta w | v \rangle = \overline{\langle v | \alpha u + \beta w \rangle} = \overline{\alpha \langle v | u \rangle + \beta \langle v | w \rangle} = \overline{\alpha} \overline{\langle v | u \rangle} + \overline{\beta} \overline{\langle v | w \rangle} = \overline{\alpha} \langle u | v \rangle + \overline{\beta} \langle w | v \rangle$ . (iv)  $\langle v | v \rangle = \sum_i \overline{v_i} v_i = \sum_i |v_i|^2 \geq 0$ , real, and the sum of nonnegative terms is 0 iff every  $|v_i|^2 = 0$ , i.e.  $v = \mathbf{0}$ .  $\square$

## 4.3 Norm

**Definition 4.2** (Induced norm).  $\|v\| = \sqrt{\langle v | v \rangle} = \sqrt{\sum_{i=0}^{n-1} |v_i|^2} \in \mathbb{R}_{\geq 0}$ .

Scaling pulls out the modulus, using (iii) then (ii):

$$\|\alpha v\|^2 = \langle \alpha v | \alpha v \rangle = \overline{\alpha} \alpha \langle v | v \rangle = |\alpha|^2 \|v\|^2 \implies \|\alpha v\| = |\alpha| \|v\|. \quad (36)$$

A vector with  $\|v\| = 1$  is a *unit vector*; quantum states will be exactly these.

## 4.4 Orthogonality and orthonormal bases

**Definition 4.3** (Orthogonality).  $u$  and  $v$  are *orthogonal*, written  $u \perp v$ , iff  $\langle u | v \rangle = 0$ . A set is *orthonormal* iff its vectors are pairwise orthogonal and each has norm 1.

Write the *Kronecker delta*  $\delta_{ij} = 1$  if  $i = j$  and 0 otherwise. The standard basis is orthonormal, because  $(e_i)_k = \delta_{ik}$  gives

$$\langle e_i | e_j \rangle = \sum_k \overline{(e_i)_k} (e_j)_k = \sum_k \delta_{ik} \delta_{jk} = \delta_{ij}. \quad (37)$$

**Coordinates are inner products.** For an orthonormal basis, the  $i$ -th coordinate of  $v$  is just its overlap with  $e_i$ . Using (26) and linearity in the second slot,

$$\langle e_i | v \rangle = \left\langle e_i \left| \sum_j v_j e_j \right. \right\rangle = \sum_j v_j \langle e_i | e_j \rangle = \sum_j v_j \delta_{ij} = v_i. \quad (38)$$

This single line — *coordinate = inner product with a basis vector* — is the seed of the Born rule: a measurement amplitude is exactly such an overlap.

#### 4.5 Cauchy–Schwarz and the triangle inequality

**Theorem 4.2** (Cauchy–Schwarz). *For all  $u, v \in \mathbb{C}^n$ ,  $|\langle u | v \rangle| \leq \|u\| \|v\|$ , with equality iff  $u, v$  are linearly dependent.*

*Proof.* If  $v = \mathbf{0}$  both sides are 0. Otherwise  $\langle v | v \rangle > 0$ ; set  $c = \frac{\langle v | u \rangle}{\langle v | v \rangle}$  and  $w = u - cv$  (fig. 5). First,  $w$  is orthogonal to  $v$  in the second slot:

$$\langle v | w \rangle = \langle v | u \rangle - c \langle v | v \rangle = \langle v | u \rangle - \langle v | u \rangle = 0 \Rightarrow \langle w | v \rangle = \overline{\langle v | w \rangle} = 0.$$

Now expand  $\langle w | w \rangle \geq 0$ , using  $\langle w | v \rangle = 0$  and conjugate-linearity in the first slot (so the scalar  $c$  pulled from the first slot becomes  $\bar{c}$ ):

$$0 \leq \langle w | w \rangle = \langle u - cv | u - cv \rangle = \langle u | u \rangle - \bar{c} \langle v | u \rangle. \quad (39)$$

With  $\bar{c} = \frac{\overline{\langle v | u \rangle}}{\langle v | v \rangle} = \frac{\langle u | v \rangle}{\|v\|^2}$  and  $\langle u | v \rangle \langle v | u \rangle = \langle u | v \rangle \overline{\langle u | v \rangle} = |\langle u | v \rangle|^2$ ,

$$0 \leq \|u\|^2 - \frac{|\langle u | v \rangle|^2}{\|v\|^2} \implies |\langle u | v \rangle|^2 \leq \|u\|^2 \|v\|^2. \quad (40)$$

Take square roots. Equality holds iff  $\langle w | w \rangle = 0$  iff  $w = \mathbf{0}$  iff  $u = cv$ .  $\square$

**Theorem 4.3** (Triangle inequality).  $\|u + v\| \leq \|u\| + \|v\|$ .

*Proof.* Expand and use  $\langle u | v \rangle + \langle v | u \rangle = \langle u | v \rangle + \overline{\langle u | v \rangle} = 2 \operatorname{Re} \langle u | v \rangle$ :

$$\|u + v\|^2 = \langle u + v | u + v \rangle = \langle u | u \rangle + \langle u | v \rangle + \langle v | u \rangle + \langle v | v \rangle = \|u\|^2 + 2 \operatorname{Re} \langle u | v \rangle + \|v\|^2 \quad (41)$$

$$\leq \|u\|^2 + 2|\langle u | v \rangle| + \|v\|^2 \leq \|u\|^2 + 2\|u\|\|v\| + \|v\|^2 = (\|u\| + \|v\|)^2, \quad (42)$$

using  $\operatorname{Re} z \leq |z|$  then Cauchy–Schwarz. Take square roots.  $\square$

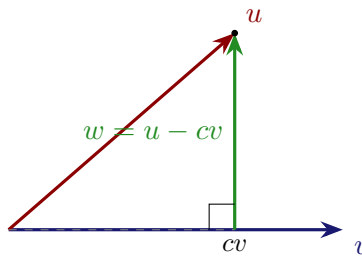


Figure 5: Geometry behind Cauchy–Schwarz. Subtracting from  $u$  its component along  $v$ , with  $c = \langle v | u \rangle / \langle v | v \rangle$ , leaves a residual  $w = u - cv$  orthogonal to  $v$ . Since  $\langle w | w \rangle \geq 0$ , expanding this inequality yields  $|\langle u | v \rangle| \leq \|u\| \|v\|$ , with equality exactly when  $w = \mathbf{0}$ , i.e.  $u$  and  $v$  are parallel.

## 4.6 Worked example

**Example 4.1.** In  $\mathbb{C}^2$  take  $u = \begin{pmatrix} 1 \\ i \end{pmatrix}$ ,  $v = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ .

**Inner products** (conjugate the *first* slot):

$$\langle u | v \rangle = \bar{1} \cdot 1 + \bar{i} \cdot 1 = 1 + (-i) = 1 - i, \quad (43)$$

$$\langle v | u \rangle = \bar{1} \cdot 1 + \bar{1} \cdot i = 1 + i = \overline{\langle u | v \rangle}. \quad \checkmark \text{ (conjugate symmetry)} \quad (44)$$

**Norms.**  $\|u\|^2 = |1|^2 + |i|^2 = 2$ ,  $\|v\|^2 = |1|^2 + |1|^2 = 2$ , so  $\|u\| = \|v\| = \sqrt{2}$ .

**Cauchy–Schwarz check.**  $|\langle u | v \rangle| = |1 - i| = \sqrt{2} \leq \|u\| \|v\| = \sqrt{2} \cdot \sqrt{2} = 2$ .  $\checkmark$  (strict, since  $u \nparallel v$ ).

**Coordinate as overlap.** With  $e_0 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ :  $\langle e_0 | v \rangle = \bar{1} \cdot 1 + \bar{0} \cdot 1 = 1 = v_0$ , matching (38).  $\checkmark$

## 4.7 Why this matters for quantum computing

**Remark 4.1.** • **States are unit vectors:**  $\|\psi\| = 1$  is the normalization condition, so total probability is 1.

- **Overlap is probability amplitude:**  $\langle \phi | \psi \rangle$  measures how much  $\psi$  “contains”  $\phi$ ; the Born rule reads off  $|\langle \phi | \psi \rangle|^2$  as a probability, and Cauchy–Schwarz keeps it  $\leq 1$  for unit vectors.
- **Orthonormal = distinguishable:** an orthonormal basis is a set of perfectly distinguishable measurement outcomes, with amplitudes  $\langle e_i | \psi \rangle = \psi_i$  by (38).

Next (§5): *Dirac notation* — repackage vectors as kets  $|\psi\rangle$  and the conjugate-linear “first slot” as bras  $\langle\psi|$ , so that  $\langle\phi | \psi\rangle$  becomes a product of a bra and a ket. It is a change of notation, not of mathematics.

## 5 Dirac notation

Nothing new happens to the mathematics here: Dirac notation is a relabelling of the vectors, functionals, and inner products of §3–§4 into a form where operator algebra is almost automatic. The payoff is the *outer product* and the *completeness relation*, which we use constantly later.

### 5.1 Kets, bras, and their shapes

**Definition 5.1** (Ket and bra). A *ket*  $|\psi\rangle$  is a column vector in  $\mathbb{C}^n$  (shape  $n \times 1$ ). Its *bra*  $\langle\psi|$  is the conjugate transpose — the adjoint — of that column, a row vector (shape  $1 \times n$ ):

$$|\psi\rangle = \begin{pmatrix} \psi_0 \\ \vdots \\ \psi_{n-1} \end{pmatrix}, \quad \langle\psi| = |\psi\rangle^\dagger = (\bar{\psi}_0 \ \cdots \ \bar{\psi}_{n-1}).$$

**Flag.** The label inside a ket is just a *name*:  $|0\rangle$  is a specific basis vector,  $|\psi\rangle$  a named vector. Arithmetic happens on kets, not inside the label — write  $|\phi\rangle + |\psi\rangle$ , never “ $|\phi + \psi\rangle$ ”.

The computational basis vectors  $e_i$  of §3 are renamed  $|0\rangle, \dots, |n-1\rangle$ , so the decomposition (26) reads

$$|\psi\rangle = \sum_{i=0}^{n-1} \psi_i |i\rangle, \quad \psi_i \in \mathbb{C}. \quad (45)$$

## 5.2 Bra–ket is the inner product; ket–bra is an operator

A bra is a *linear functional*: feed it a ket and a scalar drops out. Written as a matrix product,  $\langle\phi|$  ( $1 \times n$ ) times  $|\psi\rangle$  ( $n \times 1$ ) gives a  $1 \times 1$  scalar, and it is exactly the inner product of §4:

$$\langle\phi|\psi\rangle = \langle\phi|\psi\rangle = (\overline{\phi_0} \cdots \overline{\phi_{n-1}}) \begin{pmatrix} \psi_0 \\ \vdots \\ \psi_{n-1} \end{pmatrix} = \sum_{i=0}^{n-1} \overline{\phi_i} \psi_i. \quad (46)$$

This is *why* the physics convention conjugates the first slot (§4): the conjugation lives in the bra. Reversing the order multiplies  $|\psi\rangle$  ( $n \times 1$ ) by  $\langle\phi|$  ( $1 \times n$ ) and gives an  $n \times n$  matrix, the *outer product*  $|\psi\rangle\langle\phi|$ . Figure 6 is the shape bookkeeping.

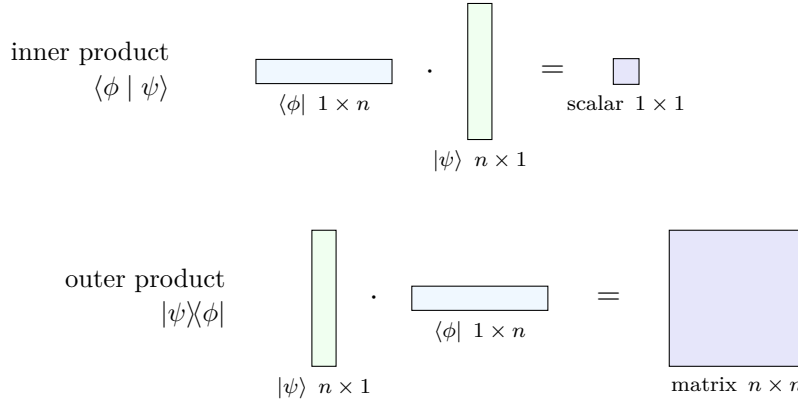


Figure 6: Bra–ket bookkeeping. A bra is a  $1 \times n$  row, a ket is an  $n \times 1$  column. **Bra**·**ket** contracts the shared dimension  $n$  down to a  $1 \times 1$  *scalar* (the inner product). **Ket**·**bra** instead expands to an  $n \times n$  *matrix* (the outer product — an operator). In both, the inner dimensions must match.

| Object  | Dirac                      | Shape        | Role                     |
|---------|----------------------------|--------------|--------------------------|
| ket     | $ \psi\rangle$             | $n \times 1$ | a vector / state         |
| bra     | $\langle\psi $             | $1 \times n$ | a linear functional      |
| bra–ket | $\langle\phi \psi\rangle$  | $1 \times 1$ | inner product (a scalar) |
| ket–bra | $ \psi\rangle\langle\phi $ | $n \times n$ | operator (outer product) |

## 5.3 The ket-to-bra map is conjugate-linear

Because the bra is an adjoint, pulling a scalar through the correspondence conjugates it:

$$(\alpha|\psi\rangle)^\dagger = \overline{\alpha}|\psi\rangle^\dagger = \overline{\alpha}\langle\psi|. \quad (47)$$

So  $\alpha|\psi\rangle \leftrightarrow \overline{\alpha}\langle\psi|$ . Concretely, if  $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle)$  then  $\langle\psi| = \frac{1}{\sqrt{2}}(\langle 0| - i\langle 1|)$  — the  $i$  flips sign.

## 5.4 Projectors and the completeness relation

The outer product  $|i\rangle\langle i|$  is the matrix with a single 1 at position  $(i, i)$ ; it *projects* onto the  $i$ -th axis. Summing them rebuilds the identity:

**Theorem 5.1** (Completeness / resolution of the identity).  $\sum_{i=0}^{n-1} |i\rangle\langle i| = \mathbf{1}$ .

*Proof.* Apply the left side to an arbitrary  $|\psi\rangle$  and use  $\psi_i = \langle i | \psi \rangle$  from (38):

$$\left( \sum_i |i\rangle\langle i| \right) |\psi\rangle = \sum_i |i\rangle \underbrace{\langle i | \psi \rangle}_{= \psi_i \in \mathbb{C}} = \sum_i \psi_i |i\rangle = |\psi\rangle.$$

(The scalar  $\langle i | \psi \rangle$  commutes freely past the ket.) Since this holds for every  $|\psi\rangle$ , the operator equals  $\mathbf{1}$ .  $\square$

Inserting  $\sum_i |i\rangle\langle i|$  anywhere (it is just  $\mathbf{1}$ ) is the standard trick for expanding a state or operator in a basis; we will reach for it repeatedly.

## 5.5 Norm in Dirac form

$\langle \psi | \psi \rangle = \sum_i |\psi_i|^2 = \|\psi\|^2$ , so a *quantum state* is a ket with  $\langle \psi | \psi \rangle = 1$ .

## 5.6 Worked example: a single qubit

**Example 5.1.** Let  $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$  in  $\mathbb{C}^2$ .

**Bra.** The amplitudes  $\frac{1}{\sqrt{2}}$  are real, so  $\langle + | = \frac{1}{\sqrt{2}}(\langle 0 | + \langle 1 |) = \frac{1}{\sqrt{2}}(1 \ 1)$ .

**Normalization.**  $\langle + | + \rangle = \frac{1}{2}(\langle 0 | 0 \rangle + \langle 0 | 1 \rangle + \langle 1 | 0 \rangle + \langle 1 | 1 \rangle) = \frac{1}{2}(1 + 0 + 0 + 1) = 1$ . So  $|+\rangle$  is a valid state.  $\checkmark$

**Amplitude and Born rule.**  $\langle 0 | + \rangle = \frac{1}{\sqrt{2}}$ , hence the probability of measurement outcome 0 is  $|\langle 0 | + \rangle|^2 = \frac{1}{2}$ .

**An outer product.**  $|0\rangle\langle 1| = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ .

**Completeness check** ( $n = 2$ ).  $|0\rangle\langle 0| + |1\rangle\langle 1| = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \mathbf{1}$ .  $\checkmark$

## 5.7 Why this matters for quantum computing

**Remark 5.1.** • **Operators read naturally:** a gate acts as  $|\psi\rangle \mapsto U|\psi\rangle$ , and  $|\phi\rangle\langle\phi|$  is the projector onto  $|\phi\rangle$  — the building block of measurement (§9).

- **Completeness is a workhorse:** inserting  $\sum_i |i\rangle\langle i| = \mathbf{1}$  expands states and operators in any orthonormal basis without changing them.
- **Amplitudes are bra-kets:**  $\langle i | \psi \rangle = \psi_i$ , and  $|\langle i | \psi \rangle|^2$  is the Born-rule probability of outcome  $i$ .

Next (§6): *linear operators* as matrices acting on kets — how a map  $\mathbb{C}^n \rightarrow \mathbb{C}^n$  is fixed by its action on a basis, and how outer products assemble any operator.

## 6 Linear operators

Quantum gates and observables are *linear operators* on  $\mathbb{C}^n$ . This step pins down what an operator is, how a matrix encodes it, and — the payoff — how outer products and the completeness relation assemble *any* operator and turn composition into matrix multiplication.



## 6.1 Definition and matrix elements

**Definition 6.1** (Linear operator). A *linear operator* is a map  $A : \mathbb{C}^n \rightarrow \mathbb{C}^n$  with  $A(\alpha|u\rangle + \beta|v\rangle) = \alpha A|u\rangle + \beta A|v\rangle$  for all  $|u\rangle, |v\rangle \in \mathbb{C}^n$  and  $\alpha, \beta \in \mathbb{C}$ .

Linearity means  $A$  is determined by its action on the basis. Writing the image of  $|j\rangle$  in coordinates,

$$A|j\rangle = \sum_{i=0}^{n-1} A_{ij} |i\rangle, \quad A_{ij} = \langle i | A | j \rangle = \langle i | A | j \rangle \in \mathbb{C}, \quad (48)$$

where the second equality uses the coordinate formula (38). The numbers  $A_{ij}$  are the *matrix elements*:  $A_{ij}$  sits in row  $i$ , column  $j$ , and column  $j$  is exactly the image  $A|j\rangle$  (fig. 7).

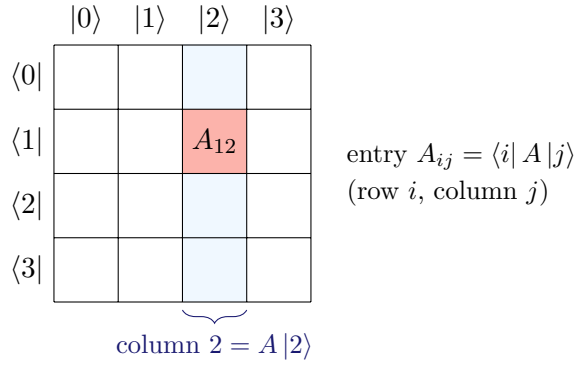


Figure 7: An operator  $A$  as a matrix in the standard basis. The entry in row  $i$ , column  $j$  is the number  $A_{ij} = \langle i | A | j \rangle$ . Column  $j$  as a whole is the coordinate vector of  $A|j\rangle$  — the image of the basis ket  $|j\rangle$ . Fixing what  $A$  does to each  $|j\rangle$  fixes the whole matrix.

Acting on a general state  $|\psi\rangle = \sum_j \psi_j |j\rangle$  reproduces the matrix–vector product (shapes:  $(n \times n)(n \times 1) = (n \times 1)$ ):

$$A|\psi\rangle = \sum_j \psi_j A|j\rangle = \sum_i \left( \sum_j A_{ij} \psi_j \right) |i\rangle. \quad (49)$$

## 6.2 Every operator is a sum of outer products

Sandwiching  $A$  between two copies of the completeness relation  $\sum_i |i\rangle\langle i| = \mathbb{1}$  (§5) expands it in outer products:

$$A = \mathbb{1} A \mathbb{1} = \left( \sum_i |i\rangle\langle i| \right) A \left( \sum_j |j\rangle\langle j| \right) \quad (50)$$

$$= \sum_{i,j} |i\rangle \underbrace{\langle i | A | j \rangle}_{= A_{ij}} \langle j| = \sum_{i,j} A_{ij} |i\rangle\langle j|. \quad (51)$$

So  $\{|i\rangle\langle j|\}$  is a basis for operators, and  $A_{ij}$  are its coordinates.

## 6.3 Composition is matrix multiplication

Insert  $\mathbb{1} = \sum_j |j\rangle\langle j|$  between two operators:

$$(AB)_{ik} = \langle i | AB | k \rangle = \langle i | A \mathbb{1} B | k \rangle = \sum_j \langle i | A | j \rangle \langle j | B | k \rangle = \sum_j A_{ij} B_{jk}. \quad (52)$$

That is the usual row-times-column rule. Crucially, operators generally **do not commute**:  $AB \neq BA$  in general (example below). Order matters, and it will matter physically — applying gates in a different order gives a different result.

## 6.4 Action on bras

Operators act to the right on kets and to the left on bras. Taking adjoints (the conjugate transpose, developed fully in §7),

$$(A|\psi\rangle)^\dagger = \langle\psi|A^\dagger, \quad (53)$$

so the bra of  $A|\psi\rangle$  involves  $A^\dagger$ , not  $A$ . We use this constantly once the adjoint is in hand.

## 6.5 Worked example: operators on a qubit

**Example 6.1.** On  $\mathbb{C}^2$  take the bit-flip  $X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$  and the phase-flip  $Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ .

**Action on the basis.**  $X|0\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} = |1\rangle$  and  $X|1\rangle = |0\rangle$ :  $X$  swaps  $|0\rangle \leftrightarrow |1\rangle$ . Its matrix elements  $X_{ij} = \langle i|X|j\rangle$  give  $X_{01} = X_{10} = 1$ ,  $X_{00} = X_{11} = 0$ , matching the matrix.

**Outer-product form** via (51):  $X = |0\rangle\langle 1| + |1\rangle\langle 0|$ .

**Non-commutativity.**

$$XZ = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad ZX = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}. \quad (54)$$

So  $XZ = -ZX \neq ZX$ : these gates *anticommute*. Order is not optional.

## 6.6 Why this matters for quantum computing

**Remark 6.1.** • **Gates are operators:** a quantum gate is a (unitary, §7) operator acting  $|\psi\rangle \mapsto U|\psi\rangle$ ; a circuit is a product of such operators, evaluated right-to-left.

- **Outer products build everything:** (51) lets us write gates and measurements directly in Dirac form, e.g. a projector  $|\phi\rangle\langle\phi|$ .
- **Order matters:** non-commutativity is not a nuisance but the source of genuinely quantum behaviour (uncertainty, interference between gate orderings).

Next (§7): the *adjoint*  $A^\dagger$ , and the two operator classes built from it — *Hermitian* ( $A^\dagger = A$ , the observables) and *unitary* ( $U^\dagger U = \mathbb{1}$ , the gates).

# 7 Adjoint, Hermitian, and unitary operators

Two operator classes carry all of quantum computing: *Hermitian* operators are the observables (their eigenvalues are the measurable numbers), and *unitary* operators are the gates (they preserve normalization, hence map states to states). Both are defined through one construction: the adjoint.

## 7.1 The adjoint

**Definition 7.1** (Adjoint). The *adjoint*  $A^\dagger$  of an operator  $A$  is the unique operator satisfying

$$\langle u | Av \rangle = \langle A^\dagger u | v \rangle \quad \text{for all } |u\rangle, |v\rangle \in \mathbb{C}^n. \quad (55)$$

In coordinates this is the *conjugate transpose*. Take  $|u\rangle = |i\rangle$ ,  $|v\rangle = |j\rangle$  in (55): the left side is  $A_{ij}$ , and the right side is  $\langle A^\dagger i | j \rangle = \overline{\langle j | A^\dagger i \rangle} = \overline{(A^\dagger)_{ji}}$ . Hence

$$A_{ij} = \overline{(A^\dagger)_{ji}} \iff (A^\dagger)_{ij} = \overline{A_{ji}} : \quad (56)$$

transpose, then conjugate every entry.

**Algebra of the adjoint.** From  $(A^\dagger)_{ij} = \overline{A_{ji}}$ :

$$(A^\dagger)^\dagger = A, \quad (\alpha A)^\dagger = \overline{\alpha} A^\dagger, \quad (A + B)^\dagger = A^\dagger + B^\dagger, \quad (AB)^\dagger = B^\dagger A^\dagger. \quad (57)$$

The order-reversal in  $(AB)^\dagger = B^\dagger A^\dagger$  comes straight from the index sum:

$$((AB)^\dagger)_{ij} = \overline{(AB)_{ji}} = \overline{\sum_k A_{jk} B_{ki}} = \sum_k \overline{B_{ki}} \overline{A_{jk}} = \sum_k (B^\dagger)_{ik} (A^\dagger)_{kj} = (B^\dagger A^\dagger)_{ij}.$$

And, as used in §6,  $(A|\psi\rangle)^\dagger = \langle\psi|A^\dagger$ .

## 7.2 Hermitian operators (observables)

**Definition 7.2** (Hermitian / self-adjoint).  $A$  is *Hermitian* iff  $A^\dagger = A$ , i.e.  $A_{ij} = \overline{A_{ji}}$  (so diagonal entries are real).

**Theorem 7.1** (Hermitian  $\Rightarrow$  real eigenvalues). If  $A^\dagger = A$  and  $A|v\rangle = \lambda|v\rangle$  with  $|v\rangle \neq \mathbf{0}$ , then  $\lambda \in \mathbb{R}$ .

*Proof.*  $\lambda\langle v | v \rangle = \langle v | Av \rangle \stackrel{(55)}{=} \langle A^\dagger v | v \rangle = \langle Av | v \rangle = \overline{\langle v | Av \rangle} = \overline{\lambda\langle v | v \rangle}$ , using  $A^\dagger = A$  then conjugate symmetry. Since  $\langle v | v \rangle > 0$ , divide to get  $\lambda = \overline{\lambda}$ , i.e.  $\lambda \in \mathbb{R}$ .  $\square$

Real eigenvalues are why Hermitian operators model measurable quantities: a measurement returns one of these real eigenvalues (§9).

## 7.3 Unitary operators (gates)

**Definition 7.3** (Unitary).  $U$  is *unitary* iff  $U^\dagger U = U U^\dagger = \mathbb{1}$ , equivalently  $U^\dagger = U^{-1}$ .

**Theorem 7.2** (Unitary  $\Leftrightarrow$  preserves the inner product).  $U^\dagger U = \mathbb{1}$  iff  $\langle Uu | Uv \rangle = \langle u | v \rangle$  for all  $|u\rangle, |v\rangle$ . In particular  $\|U\psi\| = \|\psi\|$ : norms (hence normalization) are preserved.

*Proof.* Using the adjoint relation (55) twice,

$$\langle Uu | Uv \rangle = \langle u | U^\dagger Uv \rangle.$$

If  $U^\dagger U = \mathbb{1}$  this equals  $\langle u | v \rangle$ . Conversely, if  $\langle u | U^\dagger Uv \rangle = \langle u | v \rangle = \langle u | \mathbb{1}v \rangle$  holds for all  $|u\rangle$ , then  $U^\dagger U = \mathbb{1}$  (an operator is fixed by all its inner products). Setting  $u = v$  gives  $\|Uv\|^2 = \|v\|^2$ .  $\square$

**Theorem 7.3** (Unitary  $\Rightarrow$  unit-modulus eigenvalues). If  $U|v\rangle = \lambda|v\rangle$ ,  $|v\rangle \neq \mathbf{0}$ , then  $|\lambda| = 1$ .

*Proof.*  $\langle v | v \rangle = \langle Uv | Uv \rangle = \langle \lambda v | \lambda v \rangle = \overline{\lambda}\lambda\langle v | v \rangle = |\lambda|^2\langle v | v \rangle$ , so  $|\lambda|^2 = 1$ .  $\square$

Norm preservation is exactly what a gate needs: it carries a unit state to a unit state. The columns of  $U$  are an orthonormal basis (that is  $U^\dagger U = \mathbb{1}$  read column-by-column).

## 7.4 Normal operators and the classification

**Definition 7.4** (Normal).  $A$  is *normal* iff  $AA^\dagger = A^\dagger A$ .

Both Hermitian ( $A^\dagger = A$ ) and unitary ( $U^\dagger = U^{-1}$ ) operators are normal. Normality is precisely the condition for an orthonormal eigenbasis to exist — the spectral theorem of §9. Figure 8 places the classes.

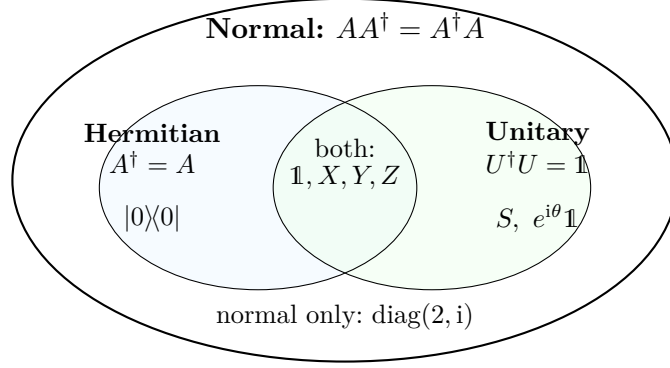


Figure 8: Operators classified by the adjoint  $A^\dagger$  (conjugate transpose). *Hermitian* ( $A^\dagger = A$ ) have *real* eigenvalues — the observables. *Unitary* ( $U^\dagger U = \mathbb{1}$ ) have unit-modulus eigenvalues and preserve the norm — the gates. Their intersection is the involutions  $A^\dagger = A$ ,  $A^2 = \mathbb{1}$  (e.g. the Paulis  $X, Y, Z$ ). All of these are *normal* ( $AA^\dagger = A^\dagger A$ ), the class the spectral theorem (§9) diagonalizes.

| Class     | Condition                  | Eigenvalues                      | Role                   |
|-----------|----------------------------|----------------------------------|------------------------|
| Hermitian | $A^\dagger = A$            | real ( $\subset \mathbb{R}$ )    | observables            |
| Unitary   | $U^\dagger U = \mathbb{1}$ | unit modulus ( $ \lambda  = 1$ ) | gates                  |
| Normal    | $AA^\dagger = A^\dagger A$ | (complex)                        | spectral-theorem class |

## 7.5 Worked example: Pauli operators and a phase gate

**Example 7.1.** The Pauli matrices and the phase gate  $S$ :

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}.$$

**$Y$  is Hermitian.**  $(Y^\dagger)_{12} = \overline{Y_{21}} = \overline{-i} = i = Y_{12}$  and  $(Y^\dagger)_{21} = \overline{Y_{12}} = \overline{i} = -i = Y_{21}$ , so  $Y^\dagger = Y$ . (Likewise  $X, Z$  are Hermitian: real and symmetric.)

**$X, Y, Z$  are unitary** because each squares to the identity, e.g.  $X^\dagger X = X^2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}^2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \mathbb{1}$ . Being Hermitian *and* unitary, they sit in the intersection of fig. 8; their eigenvalues are  $\pm 1$  (real *and* unit modulus).

**$S$  is unitary but not Hermitian.**  $S^\dagger = \begin{pmatrix} 1 & 0 \\ 0 & -i \end{pmatrix}$ , so  $S^\dagger S = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \mathbb{1}$  (unitary), yet  $S^\dagger \neq S$  (not Hermitian). Its eigenvalues  $1, i$  have modulus 1 but  $i \notin \mathbb{R}$ .

## 7.6 Why this matters for quantum computing

**Remark 7.1.** • **Gates = unitary:** reversibility ( $U^{-1} = U^\dagger$ ) and norm preservation are the defining requirements of quantum evolution.

- **Observables = Hermitian:** measurement outcomes are the (real) eigenvalues; the measurement statistics come from the eigenvectors (§9).
- **Normal = diagonalizable:** both classes are normal, so the spectral theorem will give each an orthonormal eigenbasis — the tool that makes gates and measurements computable.

Next (§8): the *tensor product*  $\otimes$ , which combines several qubits' spaces into the single space  $\mathbb{C}^{2^n}$  where multi-qubit states and entanglement live.

## 8 The tensor product $\otimes$

A single qubit lives in  $\mathbb{C}^2$ . Two qubits do *not* live in  $\mathbb{C}^2 \times \mathbb{C}^2$ ; they share one vector in  $\mathbb{C}^2 \otimes \mathbb{C}^2 \cong \mathbb{C}^4$ . The *tensor product* is the operation that builds this joint space, and it is where the state space turns exponential — and where entanglement appears.

### 8.1 Tensor product of vectors (Kronecker product)

**Definition 8.1** (Tensor product of vectors). For  $|a\rangle \in \mathbb{C}^m$  and  $|b\rangle \in \mathbb{C}^n$ , the tensor product  $|a\rangle \otimes |b\rangle \in \mathbb{C}^{mn}$  is the block (Kronecker) column

$$|a\rangle \otimes |b\rangle = \begin{pmatrix} a_0 |b\rangle \\ a_1 |b\rangle \\ \vdots \\ a_{m-1} |b\rangle \end{pmatrix}, \quad \text{entrywise} \quad (|a\rangle \otimes |b\rangle)_{in+j} = a_i b_j.$$

Figure 9 shows the  $m = n = 2$  case.

$$\begin{pmatrix} a_0 \\ a_1 \end{pmatrix} \otimes \begin{pmatrix} b_0 \\ b_1 \end{pmatrix} = \begin{pmatrix} a_0 b_0 \\ a_0 b_1 \\ a_1 b_0 \\ a_1 b_1 \end{pmatrix} \begin{array}{l} |00\rangle = |0\rangle \otimes |0\rangle \rightarrow e_0 \\ |01\rangle = |0\rangle \otimes |1\rangle \rightarrow e_1 \\ |10\rangle = |1\rangle \otimes |0\rangle \rightarrow e_2 \\ |11\rangle = |1\rangle \otimes |1\rangle \rightarrow e_3 \end{array}$$

Figure 9: Tensor product of two qubit vectors as a Kronecker (block) column. The entry at index  $2i + j$  is the product  $a_i b_j$ . The product basis  $|ij\rangle = |i\rangle \otimes |j\rangle$  maps to the standard basis  $e_{2i+j}$  of  $\mathbb{C}^4$ , so two qubits share one vector in  $\mathbb{C}^2 \otimes \mathbb{C}^2 \cong \mathbb{C}^4$ .

**Product basis.** The  $mn$  vectors  $|i\rangle \otimes |j\rangle$  ( $i < m, j < n$ ) form a basis of  $\mathbb{C}^{mn}$ . We abbreviate  $|i\rangle \otimes |j\rangle = |i\rangle |j\rangle = |ij\rangle$ . For two qubits the basis is  $|00\rangle, |01\rangle, |10\rangle, |11\rangle$ , identified with  $e_0, e_1, e_2, e_3$  of  $\mathbb{C}^4$  via the binary value  $2i + j$ .

**Bilinearity — and a flag.** The tensor product is linear in *each* slot separately:

$$(\alpha |a\rangle) \otimes |b\rangle = \alpha(|a\rangle \otimes |b\rangle) = |a\rangle \otimes (\alpha |b\rangle), \quad (|a\rangle + |a'\rangle) \otimes |b\rangle = |a\rangle \otimes |b\rangle + |a'\rangle \otimes |b\rangle, \quad (58)$$

and likewise in the second slot. **Flag:** it is *bilinear*, not linear on the pair — in general  $|a\rangle \otimes |b\rangle + |a'\rangle \otimes |b'\rangle$  does *not* collapse to a single product, which is exactly what makes entanglement possible.

## 8.2 Inner product and dimension

Inner products multiply across the factors:

$$\langle a \otimes b | c \otimes d \rangle = \sum_{i,j} \overline{a_i b_j} c_i d_j = \left( \sum_i \overline{a_i} c_i \right) \left( \sum_j \overline{b_j} d_j \right) = \langle a | c \rangle \langle b | d \rangle. \quad (59)$$

So the product basis is orthonormal:  $\langle ij | kl \rangle = \langle i | k \rangle \langle j | l \rangle = \delta_{ik} \delta_{jl}$ . Dimensions *multiply*:

$$\dim_{\mathbb{C}}(\mathbb{C}^m \otimes \mathbb{C}^n) = mn, \quad \text{so} \quad \underbrace{\mathbb{C}^2 \otimes \dots \otimes \mathbb{C}^2}_{n \text{ qubits}} = \mathbb{C}^{2^n}. \quad (60)$$

That  $2^n$  — exponential in the qubit count — is the dimension we flagged back in §3 as the headline number.

## 8.3 Tensor product of operators

**Definition 8.2** (Tensor product of operators).  $A \otimes B$  (shape  $mn \times mn$ ) acts factor-wise:

$$(A \otimes B)(|a\rangle \otimes |b\rangle) = (A|a\rangle) \otimes (B|b\rangle), \quad (A \otimes B)_{in+j, kn+l} = A_{ik} B_{jl}. \quad (61)$$

A gate on qubit 1 alone is  $A \otimes \mathbb{1}$ ; on qubit 2 alone,  $\mathbb{1} \otimes B$ . This is how single-qubit gates act inside a multi-qubit register.

## 8.4 Product states vs. entanglement

**Definition 8.3** (Product / separable vs. entangled). A state of  $\mathbb{C}^m \otimes \mathbb{C}^n$  is a *product* (separable) state if it equals  $|a\rangle \otimes |b\rangle$  for some  $|a\rangle, |b\rangle$ . Otherwise it is *entangled*.

Not every state factors. The canonical example is the *Bell state*

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle). \quad (62)$$

**Theorem 8.1** ( $|\Phi^+\rangle$  is entangled).  $|\Phi^+\rangle$  is not a product state.

*Proof.* Suppose  $|\Phi^+\rangle = (\alpha|0\rangle + \beta|1\rangle) \otimes (\gamma|0\rangle + \delta|1\rangle)$ . Expanding by bilinearity,

$$= \alpha\gamma|00\rangle + \alpha\delta|01\rangle + \beta\gamma|10\rangle + \beta\delta|11\rangle.$$

Matching coefficients against  $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$  requires

$$\alpha\gamma = \frac{1}{\sqrt{2}}, \quad \beta\delta = \frac{1}{\sqrt{2}}, \quad \alpha\delta = 0, \quad \beta\gamma = 0.$$

From  $\alpha\delta = 0$ , either  $\alpha = 0$  (forcing  $\alpha\gamma = 0 \neq \frac{1}{\sqrt{2}}$ ) or  $\delta = 0$  (forcing  $\beta\delta = 0 \neq \frac{1}{\sqrt{2}}$ ). Both are contradictions, so no such factorization exists.  $|\Phi^+\rangle$  is entangled.  $\square$

## 8.5 Worked example: a product state for contrast

**Example 8.1.** Apply  $H \otimes \mathbb{1}$  to  $|00\rangle$ , where  $H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$  is the Hadamard gate. Factor-wise,

$$(H \otimes \mathbb{1}) |00\rangle = (H |0\rangle) \otimes (\mathbb{1} |0\rangle) = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |0\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle).$$

This *is* a product state (it factors as shown), so it is *not* entangled — in contrast to  $|\Phi^+\rangle$ , which differs only by which second-qubit basis vector accompanies  $|1\rangle$ . Entanglement is created by *two-qubit* gates (e.g. CNOT), not by single-qubit gates acting through  $\otimes$ .

## 8.6 Why this matters for quantum computing

**Remark 8.1.** • **Exponential space:**  $n$  qubits share one vector in  $\mathbb{C}^{2^n}$ ; the classical description grows exponentially, which is both the promise and the simulation cost of quantum systems.

- **Entanglement is non-separability:** states like  $|\Phi^+\rangle$  have no independent per-qubit description; it is the resource behind teleportation and quantum speedups.
- **Local gates, global state:** single-qubit gates enter as  $A \otimes \mathbb{1}$ , but two-qubit gates reach across the  $\otimes$  to create entanglement.

Next (§9): the *spectral theorem* and *projectors* — diagonalizing normal operators in an orthonormal eigenbasis, which turns Hermitian observables and unitary gates into their eigenvalue/eigenvector data and completes the bridge to the measurement postulate.

# 9 Spectral theorem and projectors

The spectral theorem diagonalizes every normal operator in an orthonormal eigenbasis, rewriting it as a weighted sum of *projectors*. This is the form in which a Hermitian observable yields measurement outcomes (with Born-rule probabilities) and a unitary gate acts as phases on its eigenvectors — the last piece before the bridge to the postulates.

## 9.1 Eigenvalues, eigenvectors, and the theorem

**Definition 9.1** (Eigenpair).  $(\lambda, |v\rangle)$  with  $|v\rangle \neq \mathbf{0}$  is an *eigenpair* of  $A$  if  $A|v\rangle = \lambda|v\rangle$ ;  $\lambda \in \mathbb{C}$  is an *eigenvalue*,  $|v\rangle$  an *eigenvector*.

**Theorem 9.1** (Spectral theorem, finite-dimensional).  $A$  is normal ( $AA^\dagger = A^\dagger A$ ) iff it has an orthonormal basis of eigenvectors  $\{|v_k\rangle\}_{k=0}^{n-1}$  with eigenvalues  $\lambda_k$ . Equivalently,

$$A = \sum_k \lambda_k |v_k\rangle\langle v_k| = \sum_k \lambda_k P_k, \quad P_k := |v_k\rangle\langle v_k|. \quad (63)$$

We take the existence of such a basis as the standard finite-dimensional spectral theorem; below we verify the structure it produces and use it. (Hermitian and unitary operators are normal, §7, so both are covered.)

## 9.2 Projectors and their algebra

Each  $P_k = |v_k\rangle\langle v_k|$  is an *orthogonal projector*. From orthonormality  $\langle v_k | v_l \rangle = \delta_{kl}$ :

$$P_k^\dagger = (|v_k\rangle\langle v_k|)^\dagger = |v_k\rangle\langle v_k| = P_k \quad (\text{Hermitian}), \quad (64)$$

$$P_k P_l = |v_k\rangle \underbrace{\langle v_k | v_l \rangle}_{\delta_{kl}} \langle v_l| = \delta_{kl} P_k \quad (\text{orthogonal; idempotent } P_k^2 = P_k), \quad (65)$$

$$\sum_k P_k = \sum_k |v_k\rangle\langle v_k| = \mathbb{1} \quad (\text{completeness in the eigenbasis}). \quad (66)$$

The last line is the completeness relation of §5, now in the orthonormal eigenbasis  $\{|v_k\rangle\}$  rather than the computational basis. Check that (63) reproduces  $A$  on an eigenvector:  $A|v_l\rangle = \sum_k \lambda_k P_k |v_l\rangle = \sum_k \lambda_k \delta_{kl} |v_l\rangle = \lambda_l |v_l\rangle$ . ✓

## 9.3 Functional calculus

Because the  $P_k$  are orthogonal and sum to  $\mathbb{1}$ , any power collapses cleanly:  $A^2 = \sum_{k,l} \lambda_k \lambda_l P_k P_l = \sum_k \lambda_k^2 P_k$ , and inductively  $A^m = \sum_k \lambda_k^m P_k$ . Hence for any function  $f$  defined on the eigenvalues,

$$f(A) = \sum_k f(\lambda_k) P_k. \quad (67)$$

This is how a gate arises as the exponential of a Hermitian generator: if  $H^\dagger = H$  with  $H = \sum_k \lambda_k P_k$ , then  $e^{iH} = \sum_k e^{i\lambda_k} P_k$  is unitary (unit-modulus eigenvalues  $e^{i\lambda_k}$ ).

## 9.4 The two classes, spectrally

- **Hermitian observable**  $A = \sum_k \lambda_k P_k$  with  $\lambda_k \in \mathbb{R}$  (§7). A measurement returns some eigenvalue  $\lambda_k$  with probability  $\langle \psi | P_k | \psi \rangle = \|P_k | \psi \rangle\|^2 = |\langle v_k | \psi \rangle|^2$  (fig. 10); these sum to  $\|\psi\|^2 = 1$ .
- **Unitary gate**  $U = \sum_k e^{i\theta_k} P_k$ : it leaves each eigenvector fixed up to a phase  $e^{i\theta_k}$  on the unit circle.

## 9.5 Worked example: diagonalizing the Pauli $X$

**Example 9.1.**  $Z = \text{diag}(1, -1)$  is already diagonal: eigenpairs  $(+1, |0\rangle)$ ,  $(-1, |1\rangle)$ , so  $Z = (+1)|0\rangle\langle 0| + (-1)|1\rangle\langle 1|$ .

$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$  is not diagonal. Its eigenvectors are  $|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle)$ :

$$X|+\rangle = |+\rangle \quad (\lambda = +1), \quad X|-\rangle = -|-\rangle \quad (\lambda = -1).$$

The projectors are

$$P_+ = |+\rangle\langle +| = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad P_- = |-\rangle\langle -| = \frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}.$$

**Algebra checks.**  $P_+ + P_- = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \mathbb{1}$  (completeness), and  $P_+ P_- = \frac{1}{4} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} = \mathbf{0}$  (orthogonal). ✓



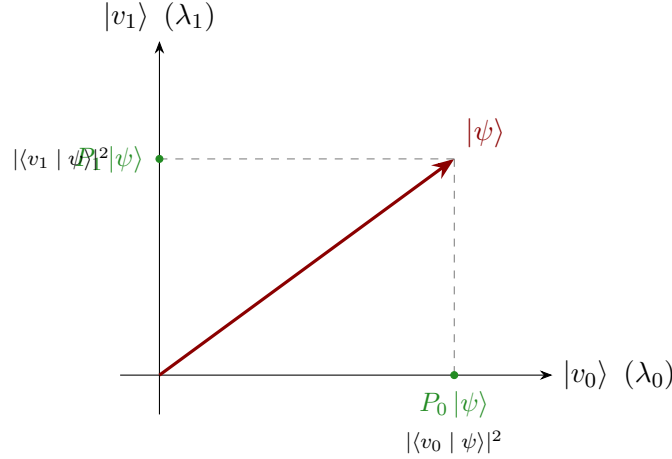


Figure 10: Spectral decomposition as measurement geometry. The orthonormal eigenvectors  $|v_k\rangle$  of a normal operator are orthogonal axes. A state  $|\psi\rangle$  splits into projections  $P_k|\psi\rangle = |v_k\rangle \langle v_k | \psi\rangle$ ; their squared lengths  $|\langle v_k | \psi\rangle|^2$  are the Born-rule probabilities of the corresponding outcomes  $\lambda_k$ , and they sum to  $\|\psi\|^2 = 1$ .

**Spectral form rebuilds  $X$ :**

$$(+1)P_+ + (-1)P_- = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} - \frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = X. \checkmark$$

**Born rule on  $|0\rangle$ .** Measuring observable  $X$  in state  $|0\rangle$ :  $|\langle + | 0\rangle|^2 = (\frac{1}{\sqrt{2}})^2 = \frac{1}{2}$  for outcome  $+1$ , and likewise  $\frac{1}{2}$  for  $-1$  — a fair coin, summing to 1.

## 9.6 Why this matters for quantum computing

**Remark 9.1.** • **Measurement is spectral:** an observable's outcomes are its eigenvalues, and the probabilities are the squared projections  $|\langle v_k | \psi\rangle|^2$  — the Born rule, read straight off (63).

- **Gates are phases in an eigenbasis:**  $U = \sum_k e^{i\theta_k} P_k$ , and  $e^{iH}$  turns any Hermitian generator into a gate via (67).
- **Diagonalization is the computational tool:** powers, exponentials, and probabilities all reduce to per-eigenvalue scalars once  $A$  is in spectral form.

That completes the math-first chain of fig. 1. The green bridge is now one short step: a *state* is a unit vector, a *gate* is a unitary, an *observable* is Hermitian, and *measurement* reads its spectral projectors via the Born rule. We can assemble the postulates whenever you want to cross over.

## 10 The bridge: postulates of quantum computing

Everything in §2–§9 was linear algebra. The *postulates* are the short dictionary that gives that algebra physical meaning. Each postulate is a naming, not a new computation: it points at an object we already built and declares what it represents. We work in  $\mathcal{H} = \mathbb{C}^{2^n}$  for  $n$  qubits.

### 10.1 Postulate 1 — States

A quantum state is a *unit vector*  $|\psi\rangle \in \mathcal{H}$ , i.e.  $\langle\psi|\psi\rangle = 1$ .

This is the norm and unit vector of §4 in the space  $\mathbb{C}^{2^n}$  of §3/§8. The amplitudes  $\psi_i = \langle i|\psi\rangle$  (§5) are the coordinates, and normalization  $\sum_i |\psi_i|^2 = 1$  is exactly “probabilities sum to 1”.

### 10.2 Postulate 2 — Composition

The state space of a composite system is the *tensor product* of the parts:  $\mathcal{H} = \mathcal{H}_1 \otimes \cdots \otimes \mathcal{H}_n = (\mathbb{C}^2)^{\otimes n} = \mathbb{C}^{2^n}$ .

This is §8 verbatim. Product states are separable; the rest are entangled (e.g.  $|\Phi^+\rangle$ ). The exponential dimension  $2^n$  is the headline number from §3.

### 10.3 Postulate 3 — Evolution (gates)

A closed system evolves by a *unitary* operator:  $|\psi\rangle \mapsto U|\psi\rangle$  with  $U^\dagger U = \mathbf{1}$ . Quantum gates are such unitaries.

This is §6 (operators acting on kets) restricted to the unitary class of §7: reversible ( $U^{-1} = U^\dagger$ ) and norm-preserving, so a state maps to a state. Continuous time is the special case  $U = e^{-iHt}$  for a Hermitian *Hamiltonian*  $H$ , built by the functional calculus of §9 ( $e^{-iHt} = \sum_k e^{-i\lambda_k t} P_k$ ).

**Overloaded-notation flag.** Italic  $H$  here is the *Hamiltonian* (a Hermitian operator); the upright sans-serif  $H$  is the *Hadamard gate*. Two different objects, similar glyphs — we keep them typographically distinct.

### 10.4 Postulate 4 — Measurement (Born rule)

An observable is a *Hermitian* operator  $A = \sum_k \lambda_k P_k$  (§9). Measuring  $A$  in state  $|\psi\rangle$  returns eigenvalue  $\lambda_k$  with probability

$$p_k = \langle\psi|P_k|\psi\rangle = \|P_k|\psi\rangle\|^2 = |\langle v_k|\psi\rangle|^2,$$

after which the state *collapses* to  $P_k|\psi\rangle/\|P_k|\psi\rangle\|$ .

The probabilities are the squared projections of fig. 10; they are real, nonnegative, and sum to  $\sum_k p_k = \langle\psi|(\sum_k P_k)|\psi\rangle = \langle\psi|\psi\rangle = 1$  by completeness (§9). *Computational-basis measurement* is the case  $A = \sum_i i|i\rangle\langle i|$ : outcome  $i$  with probability  $|\langle i|\psi\rangle|^2 = |\psi_i|^2$ .

### 10.5 Global phase is unobservable (the promise from §2)

We promised in §2 to make this precise. Replace  $|\psi\rangle$  by  $e^{i\theta}|\psi\rangle$  (a global phase). Every Born probability is unchanged:

$$\langle\psi|e^{-i\theta}P_k e^{i\theta}|\psi\rangle = e^{-i\theta}e^{i\theta}\langle\psi|P_k|\psi\rangle = \langle\psi|P_k|\psi\rangle. \quad (68)$$

So  $e^{i\theta}|\psi\rangle$  and  $|\psi\rangle$  are physically identical: a state is really an equivalence class (a *ray*)  $|\psi\rangle \sim e^{i\theta}|\psi\rangle$ . *Relative* phases between amplitudes, by contrast, do change probabilities and are observable.

## 10.6 The dictionary

| Postulate | Physical notion  | Math object                                                | Built in |
|-----------|------------------|------------------------------------------------------------|----------|
| 1         | state            | unit vector $ \psi\rangle$ , $\langle\psi \psi\rangle = 1$ | §3, §4   |
| 2         | composite system | tensor product $\mathbb{C}^{2^n}$                          | §8       |
| 3         | gate / evolution | unitary $U$ , $U^\dagger U = \mathbb{1}$                   | §6, §7   |
| 4         | observable       | Hermitian $A = \sum_k \lambda_k P_k$                       | §7, §9   |
| 4         | measurement      | Born rule $p_k =  \langle v_k   \psi \rangle ^2$           | §4, §9   |

## 10.7 Worked example: making and measuring a Bell state

**Example 10.1.** This circuit (fig. 11) exercises every postulate. Gate order is right-to-left as operators, left-to-right as drawn.

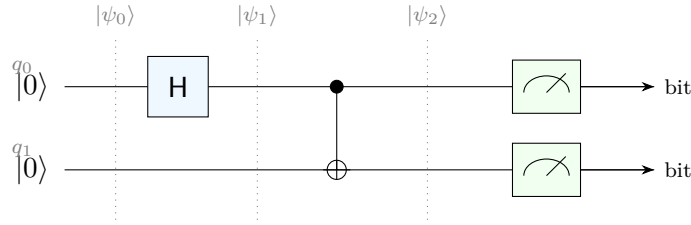


Figure 11: The Bell-state circuit, read left to right. From  $|\psi_0\rangle = |00\rangle$ : the Hadamard  $H$  on  $q_0$  gives  $|\psi_1\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$ ; the CNOT (control  $q_0$ , target  $q_1$ ) gives the entangled  $|\psi_2\rangle = |\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ ; the meters measure in the computational basis, yielding 00 or 11 with probability  $\frac{1}{2}$  each — perfectly correlated bits.

**Start (P1, P2).**  $|\psi_0\rangle = |00\rangle \in \mathbb{C}^4$ , a unit product state.

**Hadamard on  $q_0$  (P3).** With  $H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ ,

$$|\psi_1\rangle = (H \otimes \mathbb{1})|00\rangle = \left(\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)\right) \otimes |0\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle).$$

**CNOT (P3).** The controlled-NOT  $\text{CNOT} = |0\rangle\langle 0| \otimes \mathbb{1} + |1\rangle\langle 1| \otimes X$  flips  $q_1$  when  $q_0 = 1$ , so  $|00\rangle \mapsto |00\rangle$ ,  $|10\rangle \mapsto |11\rangle$ :

$$|\psi_2\rangle = \text{CNOT} |\psi_1\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = |\Phi^+\rangle.$$

By the §8 theorem this is entangled — a unitary built it from a product state.

**Measure both qubits (P4).** Projectors  $|ij\rangle\langle ij|$  give

$$p_{00} = |\langle 00 | \Phi^+ \rangle|^2 = \frac{1}{2}, \quad p_{11} = \frac{1}{2}, \quad p_{01} = p_{10} = 0,$$

collapsing to  $|00\rangle$  or  $|11\rangle$ . The two bits are random individually but *perfectly correlated* — the operational signature of entanglement.

## 10.8 Where this leads

**Remark 10.1.** The math-first chain of fig. 1 is now closed: every postulate is an object we constructed and proved things about. From here the standard next steps are all expressed in this language — single- and multi-qubit gate sets and circuits, interference and the quantum Fourier

transform (the roots of unity of §2), and algorithms (Deutsch–Jozsa, Grover, Shor). Each is a particular choice of unitaries and measurements on  $\mathbb{C}^{2^n}$ . A natural companion track is the **postgres** / **performance-tuning** project, scaffolded and waiting whenever you want to switch.

## 11 Reference card

A one-page distillation of §2–§10. Every symbol is defined in the glossary (section 12).

### State space.

$|\psi\rangle \in \mathbb{C}^{2^n}$ ,  $\langle\psi|\psi\rangle = 1$ ;  $\langle\psi| = |\psi\rangle^\dagger$ ;  
 $\langle\phi|\psi\rangle = \sum_i \phi_i \psi_i$ ;  $\|\psi\| = \sqrt{\langle\psi|\psi\rangle}$ .

### Dirac algebra.

bra·ket = scalar; ket·bra = operator.

Coordinate  $\psi_i = \langle i|\psi\rangle$ .

Completeness  $\sum_i |i\rangle\langle i| = \mathbb{1}$ .

### Operators.

$A_{ij} = \langle i|A|j\rangle$ ;  $A = \sum_{ij} A_{ij} |i\rangle\langle j|$ ;

$(AB)_{ik} = \sum_j A_{ij} B_{jk}$ ;  $(A|\psi\rangle)^\dagger = \langle\psi|A^\dagger$ .

### Classes.

Hermitian  $A^\dagger = A$  (real eig.) — observables;

Unitary  $U^\dagger U = \mathbb{1}$  ( $|\lambda| = 1$ ) — gates;

Normal  $AA^\dagger = A^\dagger A$  — diagonalizable.

### Spectral.

$A = \sum_k \lambda_k P_k$ ,  $P_k = |v_k\rangle\langle v_k|$ ;

$P_k P_l = \delta_{kl} P_k$ ,  $\sum_k P_k = \mathbb{1}$ ;

$f(A) = \sum_k f(\lambda_k) P_k$ .

### Tensor.

$(|a\rangle \otimes |b\rangle)_{in+j} = a_i b_j$ ;

$\langle a \otimes b | c \otimes d \rangle = \langle a | c \rangle \langle b | d \rangle$ ;

dim multiplies:  $n$  qubits  $\rightarrow \mathbb{C}^{2^n}$ .

Separable  $|a\rangle \otimes |b\rangle$  vs. entangled.

### Postulates.

(1) state = unit vector; (2) compose =  $\otimes$ ;

(3) gate = unitary  $U$ ; (4) observable = Hermitian,

measure: outcome  $\lambda_k$  w.p.  $p_k = |\langle v_k | \psi \rangle|^2$ ,

collapse to  $P_k |\psi\rangle / \|P_k |\psi\rangle\|$ .

Global phase  $e^{i\theta} |\psi\rangle$  unobservable.

### Computational-basis measurement.

outcome  $i$  with probability  $|\psi_i|^2$ .

| Gate         | Matrix                                                                                           | Role                                           |
|--------------|--------------------------------------------------------------------------------------------------|------------------------------------------------|
| $\mathbb{1}$ | $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$                                                   | identity                                       |
| $X$          | $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$                                                   | bit flip (NOT); Hermitian + unitary            |
| $Y$          | $\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$                                                  | Pauli-Y                                        |
| $Z$          | $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$                                                  | phase flip                                     |
| $H$          | $\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$                               | Hadamard; makes superposition                  |
| $S$          | $\begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$                                                   | phase $\frac{\pi}{2}$ ; unitary, not Hermitian |
| CNOT         | $\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$ | entangler (control $q_0$ , target $q_1$ )      |

## 12 Glossary of symbols and terms

### Symbols

$\mathbb{C}$  the complex numbers; a scalar  $z = a + ib$  with  $a, b \in \mathbb{R}$ .

$i$  the imaginary unit,  $i^2 = -1$ .

$\mathcal{H}$  a (finite-dimensional) Hilbert space; the state space of a quantum system.

$\bar{z}$  complex conjugate of  $z \in \mathbb{C}$ ; if  $z = a + ib$  then  $\bar{z} = a - ib$  (synonym:  $z^*$ ).

$\mathbb{C}^n$

the space of complex column vectors of length  $n$  (shape  $n \times 1$ ).

$A^\dagger$

adjoint of  $A$ : its conjugate transpose (“dagger”).

$\mathbb{1}$  the identity operator / matrix.

$\operatorname{Re} z, \operatorname{Im} z$

real and imaginary parts of  $z = a + ib$ ; both are *real* numbers  $a, b$ .

$|z|$

modulus of  $z$ ;  $|z| = \sqrt{a^2 + b^2} = \sqrt{z\bar{z}} \in \mathbb{R}_{\geq 0}$ .

$\arg z$

argument (angle) of  $z$  in the complex plane, defined modulo  $2\pi$ .

$e^{i\theta}$

unit-modulus phase  $\cos \theta + i \sin \theta$  (Euler's formula).

$\omega_k$

$n$ -th root of unity  $e^{2\pi i k/n}$ ,  $k \in \{0, \dots, n-1\}$ .

$v_i$  the  $i$ -th coordinate (a scalar in  $\mathbb{C}$ ) of a vector  $v \in \mathbb{C}^n$ .

$e_i$  the  $i$ -th standard (computational) basis vector of  $\mathbb{C}^n$ : a 1 in slot  $i$ , else 0.

$\mathbf{0}$  the zero vector of  $\mathbb{C}^n$  (all entries 0); distinct from the scalar 0.

$\dim_{\mathbb{C}}, \dim_{\mathbb{R}}$

complex / real dimension of a vector space;  $\dim_{\mathbb{C}} \mathbb{C}^n = n$ ,  $\dim_{\mathbb{R}} \mathbb{C}^n = 2n$ .

$\langle u | v \rangle$

Hermitian inner product  $\sum_i \bar{u}_i v_i$ ; conjugate-linear in  $u$ , linear in  $v$ .

$\|v\|$

norm  $\sqrt{\langle v | v \rangle} = \sqrt{\sum_i |v_i|^2}$ ;  $\|v\| = 1$  defines a unit vector.

$\perp$  orthogonality:  $u \perp v$  means  $\langle u | v \rangle = 0$ .

$\delta_{ij}$

Kronecker delta: 1 if  $i = j$ , else 0.

$|\psi\rangle$

ket: a column vector / state in  $\mathbb{C}^n$  (shape  $n \times 1$ ).

$\langle\psi|$

bra: the adjoint row  $|\psi\rangle^\dagger$  (shape  $1 \times n$ ); a linear functional.

$|\psi\rangle\langle\phi|$

outer product: the  $n \times n$  operator  $|\psi\rangle\langle\phi|$ .

$\sum_i |i\rangle\langle i| = \mathbb{1}$

completeness (resolution of the identity).

$A_{ij}$

matrix element  $\langle i | A | j \rangle$  of operator  $A$  (row  $i$ , column  $j$ ).

$A^\dagger$

adjoint (conjugate transpose) of  $A$ :  $(A^\dagger)_{ij} = \overline{A_{ji}}$ , defined by  $\langle u | Av \rangle = \langle A^\dagger u | v \rangle$ .

$X, Y, Z$

Pauli operators (Hermitian and unitary; each squares to  $\mathbb{1}$ , eigenvalues  $\pm 1$ ).

$S$  phase gate  $\text{diag}(1, i)$ ; unitary but not Hermitian.

$\otimes$  tensor (Kronecker) product;  $\mathbb{C}^m \otimes \mathbb{C}^n \cong \mathbb{C}^{mn}$ ,  $(|a\rangle \otimes |b\rangle)_{in+j} = a_i b_j$ .

$|ij\rangle$

product basis vector  $|i\rangle \otimes |j\rangle$ ; for qubits,  $|ij\rangle \rightarrow e_{2i+j}$ .

$|\Phi^+\rangle$

Bell state  $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ ; the canonical entangled state.

$H$  Hadamard gate  $\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ .

$\lambda_k, |v_k\rangle$

eigenvalue and (orthonormal) eigenvector:  $A|v_k\rangle = \lambda_k|v_k\rangle$ .

$P_k$

spectral projector  $|v_k\rangle\langle v_k|$ ;  $P_k P_l = \delta_{kl} P_k$ ,  $\sum_k P_k = \mathbb{1}$ .

$f(A)$

functional calculus  $\sum_k f(\lambda_k) P_k$ ; e.g.  $e^{iH} = \sum_k e^{i\lambda_k} P_k$ .

$|\pm\rangle$

Pauli- $X$  eigenstates  $\frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle)$ , eigenvalues  $\pm 1$ .

$H$  (**italic**)

Hamiltonian: a Hermitian operator generating evolution  $e^{-iHt}$  (distinct from the Hadamard  $H$ ).

CNOT

controlled-NOT  $|0\rangle\langle 0| \otimes \mathbb{1} + |1\rangle\langle 1| \otimes X$ ; flips the target qubit when control is  $|1\rangle$ .

## Terms

### finite-dimensional

every space here is  $\mathbb{C}^n$  for some finite  $n$ ; no infinite-dimensional analysis is needed.

### adjoint (dagger)

the conjugate transpose  $A^\dagger$ ; central to “Hermitian” and “unitary” later.

### modulus

the length  $|z|$  of  $z$  in the complex plane.

### argument

the angle  $\arg z$  of  $z$ ; together  $(|z|, \arg z)$  are polar coordinates.

### polar form

$z = r e^{i\theta}$  with  $r = |z|$ ,  $\theta = \arg z$ .

### Euler’s formula

$e^{i\theta} = \cos \theta + i \sin \theta$ .

**root of unity**

a solution  $\omega_k = e^{2\pi i k/n}$  of  $z^n = 1$ ; backbone of the quantum Fourier transform.

**vector space**

a set closed under vector addition and scalar multiplication, satisfying the eight axioms.

**span**

the set of all linear combinations of a given set of vectors.

**linear independence**

no nontrivial combination equals  $\mathbf{0}$ ; coordinates are then unique.

**basis**

a linearly independent spanning set; its size is the dimension.

**dimension**

the number of vectors in any basis;  $\dim_{\mathbb{C}} \mathbb{C}^n = n$ .

**standard / computational basis**

$\{e_0, \dots, e_{n-1}\}$ ; later written  $\{|0\rangle, \dots, |n-1\rangle\}$ .

**inner product (Hermitian)**

$\langle u | v \rangle = \sum_i \bar{u}_i v_i$ ; physics convention conjugates the first slot.

**norm / unit vector**

length  $\|v\|$ ; a unit vector has  $\|v\| = 1$  (a quantum state).

**orthonormal**

pairwise orthogonal, each of norm 1;  $\langle e_i | e_j \rangle = \delta_{ij}$ .

**Cauchy–Schwarz**

$|\langle u | v \rangle| \leq \|u\| \|v\|$ ; equality iff  $u, v$  are parallel.

**ket / bra**

Dirac names for a column vector  $|\psi\rangle$  and its adjoint row  $\langle\psi|$ .

**outer product**

$|\psi\rangle\langle\phi|$ , an  $n \times n$  operator;  $|i\rangle\langle i|$  projects onto axis  $i$ .

**completeness relation**

$\sum_i |i\rangle\langle i| = \mathbb{1}$ ; “insert the identity” to expand in a basis.

**linear operator**

a linear map  $A : \mathbb{C}^n \rightarrow \mathbb{C}^n$ ; an  $n \times n$  matrix once a basis is fixed.

**matrix element**

$A_{ij} = \langle i | A | j \rangle$ ; column  $j$  is the image  $A | j \rangle$ .

**non-commuting**

$AB \neq BA$  in general; e.g.  $XZ = -ZX$  for Pauli gates.

**adjoint**

conjugate transpose  $A^\dagger$ ; defined by  $\langle u | Av \rangle = \langle A^\dagger u | v \rangle$ .

**Hermitian (self-adjoint)**

$A^\dagger = A$ ; real eigenvalues; models an observable.

**unitary**

$U^\dagger U = \mathbb{1}$ ; preserves inner products and norm; models a gate.

**normal**

$AA^\dagger = A^\dagger A$ ; exactly the operators with an orthonormal eigenbasis (§9).

**tensor product**

$\otimes$ ; combines subsystems, dimensions multiply ( $n$  qubits  $\rightarrow \mathbb{C}^{2^n}$ ).

**product / separable state**

one that factors as  $|a\rangle \otimes |b\rangle$ .

**entangled state**

a joint state with no product factorization, e.g.  $|\Phi^+\rangle$ .

**spectral theorem**

every normal  $A$  has an orthonormal eigenbasis;  $A = \sum_k \lambda_k P_k$ .

**projector**

$P = P^\dagger = P^2$ ; spectral projectors  $P_k = |v_k\rangle\langle v_k|$  are orthogonal.

**Born rule**

probability of outcome  $\lambda_k$  is  $|\langle v_k | \psi \rangle|^2 = \langle \psi | P_k | \psi \rangle$ .

**ray / global phase**

a state is a class  $|\psi\rangle \sim e^{i\theta} |\psi\rangle$ ; global phase is unobservable, relative phase is not.

**postulates**

the dictionary mapping unit vector / unitary / Hermitian / Born rule to state / gate / observable / measurement.